SONOYOLO + MEDSAM2: A PIPELINE FOR AUTOMATIC DETECTION AND SEGMENTATION OF REGIONS OF INTEREST IN CARDIAC ULTRASOUND VIDEOS

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2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI) | 979-8-3315-7763-6/26/$31.00 ©2026 IEEE | DOI: 10.1109/ISBI61048.2026.11516015

ABSTRACT

Precise evaluation of cardiac function is essential for early diagnosis, risk assessment, and treatment of cardiovascular diseases. Cardiac ultrasound videos are widely used for this purpose, but cardiologists need to manually delineate the regions of interest in the videos, which is a time-consuming and error-prone process. While many automatic video segmentation methods exist, modern deep learning-based methods require dense annotations for training, which is labor-intensive. Recent zero-shot segmentation models specialized for medical data such as MedSAM2 do not require such dense annotations, but they still need manual prompts (e.g., bounding boxes), which hinders automatic deployment. To address these challenges, we propose a fully automated pipeline for detecting and segmenting regions of interest in cardiac ultrasound videos. The proposed pipeline (SonoYOLO + MedSAM2) consists of a detection model (SonoYOLO) obtained by finetuning YOLOv11-nano to localize the LV in an image, and a segmentation model (MedSAM2) that receives a bounding box as a prompt and segments the LV. The resulting image mask is used as a prompt to segment the next image and so on, thus achieving temporal consistency across the video. Experiments on EchoNet-Dynamic and EchoNet-Pediatric datasets show strong performance and generalizability of our pipeline without requiring dense annotations or manual prompts. Code is available [1].

***Index Terms*—** Semantic Segmentation, Detection, Left Ventricle, Echocardiography

1. INTRODUCTION

Cardiovascular diseases (CVDs) remain the leading causes of death worldwide, accounting for an estimated 17.9 million deaths annually [1]. Precise evaluation of cardiac function is crucial for early diagnosis, risk assessment, and treatment planning in patients with CVDs. Echocardiography is one of the most widely used screening modalities for this purpose, providing a noninvasive and cost-effective means to assess cardiac structure and function. Among the key regions of interest in echocardiography such as the left ventricle (LV), right ventricle and myocardium, the LV plays a central role in evaluating cardiac function [2]. To quantify key cardiac parameters such as LV volume, mass, and ejection fraction, cardiologists need to manually delineate LV contours in ultrasound videos, which is time-consuming and error-prone.

To overcome these limitations, automatic segmentation models [3, 4, 5] with good performance have been developed. However, most deep learning-based models require large volumes of densely annotated data for training, which are expensive to obtain. More recently, zero-shot segmentation models specialized for medical data, such as MedSAM2 [6], have been developed. While MedSAM2 does not require dense annotations, it needs user prompts (e.g., bounding box of the region of interest), which hinders automatic deployment. This motivates the development of methods that combine the generalization ability of foundation models with the efficiency of automated detection to achieve prompt-free LV segmentation.

In this paper, we propose SonoYOLO + MedSAM2, a fully automated pipeline for detecting and segmenting the LV in cardiac ultrasound videos. Our pipeline consists of a detection model (SonoYOLO) fine-tuned to localize the LV, and a segmentation model (MedSAM2) that segments the LV given either a bounding box prompt from SonoYOLO or a segmentation mask from for the previous frame from MedSAM2.

The main contributions of this work are the following:

- The proposed method is fully automatic and enables the use of MedSAM2 for echocardiographic video segmentation *without* dense annotations or manual prompts.
- The proposed method achieves an average IoU of 0.8769 for detection and a Dice Similarity Coefficient (DSC) of 0.8978 for segmentation on the EchoNet-Dynamic dataset, achieving close-to-expert performance.
- A backward propagation mechanism achieves a DSC of 0.9159, surpassing all other benchmark models.
- An evaluation on the EchoNet-Pediatric dataset, achieving a DSC of 0.8856, demonstrating strong generalizability.

2. RELATED WORK

Deep learning techniques have been widely applied in medical image segmentation in recent years to address issues such as manual dependence, inefficiency, and low scalability [7]. The emergence of convolutional neural networks (CNNs),

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[1] https://github.com/IrajShroff/SonoYOLO-MedSAM2

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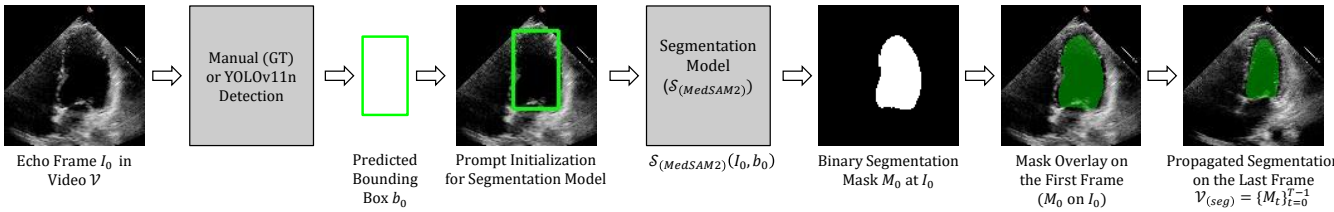


Fig. 1. High-level framework of SonoYOLO + MedSAM2. A fine-tuned YOLOv11n detector first localizes the left ventricle (LV) in the initial frame using bounding-box coordinates. The segmentation model MedSAM2 then uses this box as a spatial prompt to segment the LV and propagates the mask through subsequent frames to produce a fully segmented video.

specifically encoder-decoder architectures in computer vision, has enhanced the field of medical image segmentation [8]. Examples include models such as U-Net [9], V-Net [10], and DeepLabv3+ [11]. However, these models depend on dense pixel-level annotation and struggle across different datasets. There are also specialized models for LV segmentation such as the EchoNet-Dynamic and EchoNet-Pediatric model [3],[12]. These frameworks also rely on dense, manually annotated training masks, which limits their scalability and adaptability to new domains.

Foundation models such as SAM, SAM2, and MedSAM2 [13], [14], [6] are zero-shot segmentation models that eliminate task-specific training but require manual input prompts like bounding boxes or points, restricting their use in large-scale analysis. To overcome these limitations, we propose SonoYOLO + MedSAM2, a hybrid detection–segmentation pipeline designed for automated echocardiogram analysis.

3. METHOD

This section describes SonoYOLO + MedSAM2, a framework shown in Fig. 1 that automatically detects (Section 3.1) and segments (Section 3.2) the LV in echocardiograms.

3.1. Left Ventricle Detection

We first train a lightweight object detector to predict bounding boxes for the regions of interest. YOLOv11n (You Only Look Once) [15] is chosen because it is lightweight and efficient. This model employs a single convolutional neural network to simultaneously predict bounding boxes and class probabilities for regions of interest [16], streamlining the detection pipeline compared to more complex traditional methods [17].

SonoYOLO for LV detection. We initialize YOLOv11n with COCO-pretrained weights and fine-tune it with ultrasound data to enable it to detect the LV. Expert-annotated LV contours are provided at *end-diastole (ED)*, the frame with maximal LV cavity, and *end-systole (ES)*, the frame with minimal LV cavity, in each video. These contours are converted to tight, axis-aligned bounding boxes $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$, which are normalized and mapped to the YOLO format. To adapt YOLOv11n from natural images to echocardiograms, we fine-tune all detection layers on LV bounding boxes using the training subset of the EchoNet-Dynamic Dataset.

SonoYOLO inference. Let an echocardiogram video be represented as $\mathcal{V} = \{I_t\}_{t=0}^{T-1}$, where I_t denotes the t-th frame and T is the total number of frames. We use the SonoYOLO detector $\mathcal{D}_{\text{YOLO}}$ to localize the LV in the t-th image frame I_t by predicting a bounding box

$$\mathbf{b}_t = \mathcal{D}_{\text{YOLO}}(I_t), \tag{1}$$

where $\mathbf{b}_t = (x_{\min,t}, y_{\min,t}, x_{\max,t}, y_{\max,t})$ denotes the box coordinates normalized from the YOLO prediction.

3.2. Left Ventricle Segmentation

MedSAM2 [6] utilizes the SAM2 [14] network architecture, including an image encoder, a memory attention module, a prompt encoder, and a mask decoder. MedSAM2 can segment diverse medical data, including echocardiograms. Here, we combine MedSAM2 with SonoYOLO to segment the LV.

Specifically, after SonoYOLO predicts the bounding box coordinates $\mathbf{b}_t = \mathcal{D}_{\text{YOLO}}(I_t)$ for the t-th image frame I_t, we can use the MedSAM2 segmentation model ($\mathcal{S}_{\text{MedSAM2}}$) with $\mathbf{b}_t$ as a spatial prompt to generate the corresponding LV mask:

$$\mathbf{M}_t = \mathcal{S}_{\text{MedSAM2}}(I_t, \mathbf{b}_t). \tag{2}$$

Alternatively, we can propagate the initial mask $\mathbf{M}_0$ through the remaining frames $\{I_t\}_{t=1}^{T-1}$ by using mask $\mathbf{M}_{t-1}$ as a prompt for MedSAM2 to segment frame I_t, i.e.:

$$\mathbf{M}_t = \mathcal{S}_{\text{MedSAM2}}(I_t, \mathbf{M}_{t-1}). \tag{3}$$

The segmentation of the video is given by $\mathcal{V}_{\text{seg}} = \{\mathbf{M}_t\}_{t=0}^{T-1}$.

4. EXPERIMENTS AND RESULTS

This section demonstrate the effectiveness of the SonoYOLO + MedSAM2 detection and segmentation pipeline.

4.1. Dataset and Preprocessing

All experiments are conducted on two benchmark datasets: EchoNet-Dynamic dataset [18] and EchoNet-Pediatric dataset

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[19]. The first one is used to fine-tune the detection model and evaluate the pipeline's performance, and the second one is used to evaluate the pipeline's generalization performance.

The EchoNet-Dynamic dataset [18] includes 10,030 apical-4-chamber videos with human expert annotations (tracings and measurements). In the dataset, 7,465 videos are used to train the detector, 1,288 are used to test detection performance, and the remaining 1,277 videos are reserved to evaluate the full SonoYOLO + MedSAM2 pipeline.

To further evaluate the pipeline's generalization ability, we test the on EchoNet-Pediatric dataset [19], which includes 3,284 apical-4-chamber labeled videos. The dataset includes patients ranging from 0-18 years with a wide range of heart sizes [12]. Group 9 of the dataset, which consisted of 368 videos, is used for testing.

4.2. Experimental Setup

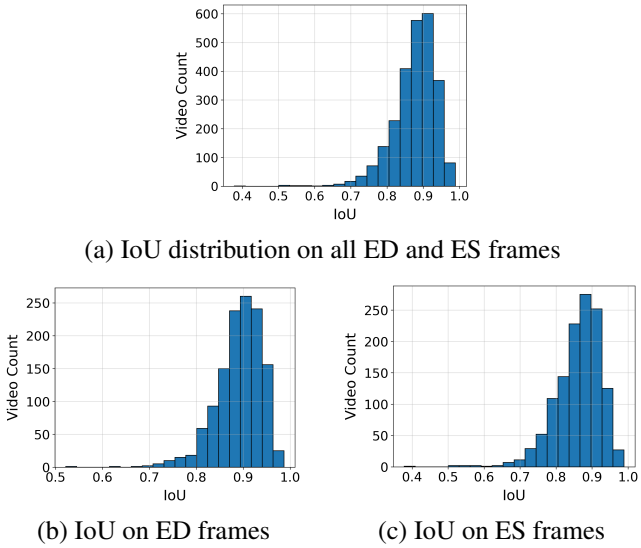
We design three experimental settings to evaluate SonoYOLO + MedSAM2 for both image- and video-level segmentations.

1. **Image (frame-wise) segmentation.** For each annotated frame I_t, which corresponds to the ED or ES frame, SonoYOLO predicts a bounding box $\mathbf{b}_t$, which we use as a spatial prompt to MedSAM2 to produce a mask $\mathbf{M}_t$. We directly compare $\mathbf{M}_t$ to the ground-truth mask $\mathbf{G}_t$ on the same frame (two evaluations per video: ED and ES).
2. **Video segmentation: forward propagation.** Starting at the first annotated frame in each video (typically ED), we obtain $\mathbf{b}_0$ with SonoYOLO and $\mathbf{M}_0$ with MedSAM2 using (1) and (2). MedSAM2 then propagates the segmentation forward by conditioning each frame's features on prior frame predictions using self-attention and cross-attention within the memory attention module, which retrieves embeddings from the streaming memory bank [6]. The predicted mask at the second annotated frame, $\mathbf{M}_T$, is then compared with the ground truth $\mathbf{G}_T$.
3. **Video segmentation: backward propagation.** Here, we start at the second annotated frame I_T (typically ES) and use SonoYOLO to obtain $\mathbf{b}_T$ and MedSAM2 to generate $\mathbf{M}_T$. MedSAM2 then propagates backward toward the first annotated frame (typically ED) in a similar way as described with forward propagation. The propagated mask at the first annotated frame, $\mathbf{M}_0$, is evaluated against $\mathbf{G}_0$.

For all MedSAM2 experiments on the EchoNet-Dynamic dataset, the detected and GT bounding boxes were uniformly shrunk by 3 pixels to removes peripheral speckle noise.

4.3. Additional Dataset Testing Setup

We evaluate zero-shot generalization ability of our pipeline on the public *EchoNet-Pediatric* dataset. For each annotated frame I_t, SonoYOLO predicts a bounding box $\mathbf{b}_t$, which is used as a spatial prompt to MedSAM2 to produce a mask $\mathbf{M}_t$. The $\mathbf{M}_t$ is compared to the GT mask $\mathbf{G}_t$ on the same frame.



(a) IoU distribution on all ED and ES frames

(b) IoU on ED frames

(c) IoU on ES frames

Fig. 2. Performance distribution of the detection model.

For EchoNet-Pediatric, no bounding box shrinkage was applied since the LV occupies a smaller proportion of the frame.

4.4. Left Ventricle Detection

The performance of SonoYOLO is shown in Fig. 2, which shows the Intersection over Union (IoU) score for the predicted bounding boxes compared to the ground truth. The model was fine-tuned for 40 epochs with a batch size of 16. The average IoU score over the annotated frames (ED, ES) in the testing subset is 0.8769. Additionally, the average IoU on ED frames is 0.8918 and the average IoU on ES frames is 0.8620. The detection IoU is higher for ED frames because in this phase, the left ventricle is largest and more clearly defined, allowing SonoYOLO to localize it more accurately.

4.5. SonoYOLO + MedSAM2 Segmentation

We evaluate the performance of SonoYOLO + MedSAM2 under the three experimental settings defined in Section 4.2: frame-by-frame segmentation (Table 1), video segmentation with forward propagation (Table 2), and video segmentation with backward propagation (Table 3). We compare against several baselines, each isolating a specific component of the pipeline: (1) EchoNet-Dynamic, the benchmark model provided by Stanford; (2) GT + SAM2 and GT + MedSAM2, uses ground-truth (GT) bounding boxes derived directly from expert contours as prompts to the segmentation models, representing higher level performance that is achieved through perfect detection; (3) SonoYOLO + SAM2, which replaces ground-truth prompts with bounding boxes predicted by our fine-tuned SonoYOLO detector, thereby assessing segmentation accuracy under automatic detection. We use the Dice

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Table 1. Segmentation performance on EchoNet-Dynamic dataset with framewise segmentation.

Method	DSC ↑	HD95 (px) ↓	MAE (%) ↓
GT + SAM2	0.8671	6.075	2.3858 ± 1.2032
SonoYOLO + SAM2	0.8525	6.458	2.6574 ± 1.3332
GT + MedSAM2	0.9083	3.886	1.5883 ± 0.6990
EchoNet-Dynamic	0.9150	N/A	N/A
SonoYOLO + MedSAM2	0.8978	4.202	1.7821 ± 0.8477

Table 2. Segmentation performance on EchoNet-Dynamic dataset with forward propagation.

Method	DSC ↑	HD95 (px) ↓	MAE (%) ↓
GT + SAM2	0.8249	8.345	2.6251 ± 1.4246
SonoYOLO + SAM2	0.8180	8.253	2.4695 ± 1.2361
GT + MedSAM2	0.8759	5.202	1.8237 ± 0.8693
SonoYOLO + MedSAM2	0.8746	5.236	1.8486 ± 0.9002

Similarity Coefficient (DSC), the 95th Percentile Hausdorff Distance (HD95) and the Mean Absolute Error (MAE) metrics to evaluate the segmentation performance. Note that while segmentation masks are propagated across all frames to produce a fully segmented video, quantitative evaluation is performed only on ED / ES frames, which are the only frames with expert annotations available in EchoNet datasets and are standard for benchmarking segmentation methods.

Framewise Segmentation. As shown in Table 1, the proposed SonoYOLO + MedSAM2 attains a DSC of 0.8978 (ED: 0.9125, ES: 0.8832), approaching the fully supervised EchoNet-Dynamic benchmark (DSC of 0.9150) despite relying on automatically detected bounding boxes rather than expert annotations, indicating robustness of MedSAM2 to moderate localization inaccuracies.

Forward Propagation. In Table 2, SonoYOLO + MedSAM2 maintains high performance under forward propagation, achieving a DSC of 0.8746, which is only 0.0013 below the GT + MedSAM2 baseline.

Backward Propagation. Backward propagation, summarized in Table 3, achieves the strongest overall results, with SonoYOLO + MedSAM2 reaching a DSC of 0.9159 and HD95 of 4.194 px, slightly surpassing the GT + MedSAM2 baseline. This improvement can be attributed to phase asymmetry across the cardiac cycle: the LV is smaller and less distinct at the ES, yielding lower framewise DSC than at the ED. By propagating from ES to the ED, where segmentation quality naturally improves, backward propagation achieves greater accuracy than forward propagation.

EchoNet-Pediatric. On this dataset, SonoYOLO + MedSAM2 achieves a DSC of 0.8856 and HD95 of 4.865 px, performing competitively with the fully supervised Echo-Peds model. This demonstrates strong generalization to pediatric ultrasound, where cardiac morphology substantially differs from the adult data used in training.

Table 3. Segmentation performance on EchoNet-Dynamic dataset with backward propagation.

Method	DSC ↑	HD95 (px) ↓	MAE (%) ↓
GT + SAM2	0.8633	7.027	2.9652 ± 1.5252
SonoYOLO + SAM2	0.8608	6.997	3.0385 ± 1.5188
GT + MedSAM2	0.9158	4.228	1.9383 ± 0.9343
SonoYOLO + MedSAM2	0.9159	4.194	1.9390 ± 0.8743

Table 4. Evaluation of segmentation performance on the EchoNet-Pediatric dataset with framewise segmentation.

Method	DSC ↑	HD95 (px) ↓	MAE (%) ↓
GT + SAM2	0.8688	6.119	2.4530 ± 1.3884
SonoYOLO + SAM2	0.8333	7.571	3.2335 ± 1.7112
GT + MedSAM2	0.9020	3.891	1.8094 ± 1.4298
Echo-Peds	0.8910	N/A	N/A
SonoYOLO + MedSAM2	0.8856	4.865	2.2621 ± 1.4287

4.6. SonoYOLO + MedSAM2 Qualitative Results

In Fig. 3, we show the qualitative results of SonoYOLO + MedSAM2, demonstrating that the model creates segmentation masks that closely align with ground-truth annotations.

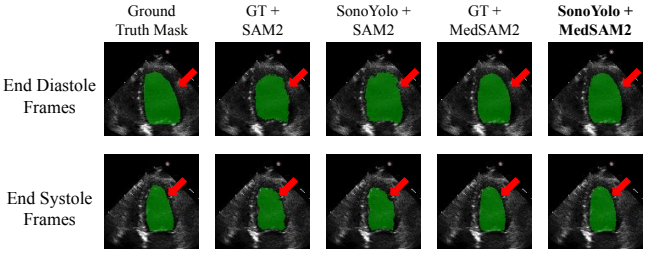


Fig. 3. **SonoYOLO + MedSAM2 qualitative results.** The first column shows the Ground Truth mask, and other columns show masks of different methods. Two rows show the End Systole and End Diastole frames, separately.

5. CONCLUSION

In this work, we introduced SonoYOLO + MedSAM2, a fully automated detection-segmentation pipeline in cardiac ultrasound videos. By combining a fine-tuned YOLOv11-nano detector with the MedSAM2 foundation model, our framework eliminates the need for manual bounding-box prompts while maintaining high segmentation accuracy. Comprehensive experiments on the EchoNet-Dynamic and EchoNet-Pediatric datasets demonstrated that the proposed pipeline achieves strong performance and robustness across diverse adult and pediatric cohorts. Future work will include evaluating the model on hard ultrasound conditions such as low-quality imaging and artifacts, to explore real-world deployment.

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6. COMPLIANCE WITH ETHICAL STANDARDS

This study used only publicly available, fully de-identified datasets: EchoNet-Dynamic and EchoNet-Pediatric datasets, which are provided by the Stanford Center for Artificial Intelligence in Medicine and Imaging (AIMI).

7. ACKNOWLEDGMENTS

Iraj sincerely acknowledges the support from Penn Engineering and the Army Educational Outreach Program. Iraj and Yuyan thank Kaleab A. Kinfu for his valuable feedback on this work. Yuyan and René acknowledge the support from the Penn Engineering Dean's Fellowship and Penn startup funds.

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